



Elastic Cloud Services for Distributed Deployment





Foreword

- In the course of business development, service systems can become overloaded, slow down, or even face interruptions due to demand fluctuations and market changes. Building scalable systems helps enhance service stability, improve response speed, and optimize resource utilization.
- This chapter describes what scalability is and how to build scalable systems on Huawei Cloud.



Objectives

- Upon completion of this lesson, you will:
 - Learn the advantages of load balancing on the cloud.
 - Learn how to schedule distributed applications.
 - Learn how to adjust resources for distributed applications.



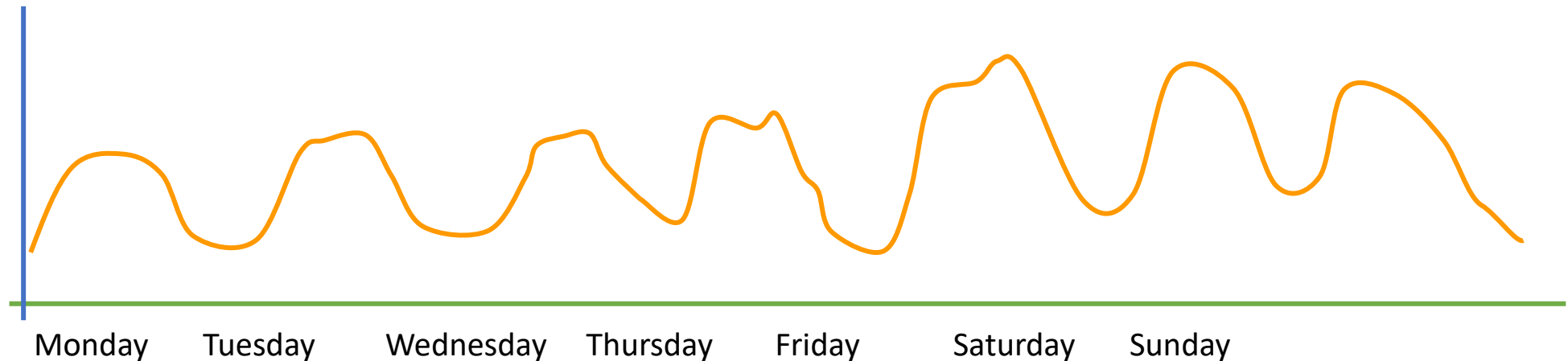
Contents

- 1. Scalability Implementation**
2. How to Build Scalable Applications
3. Typical Scalable Website Architecture

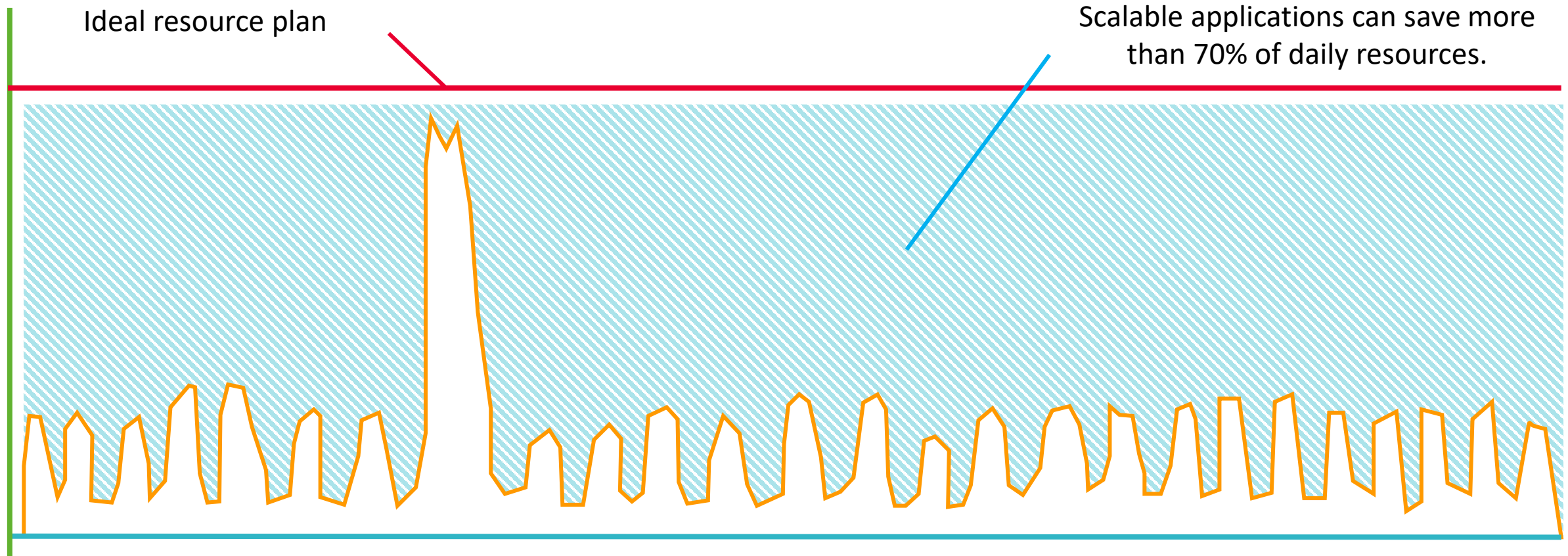
Why Is Scalability Important?

The amount of traffic an application has to handle may fluctuate.

Resources should be adjusted as traffic changes.

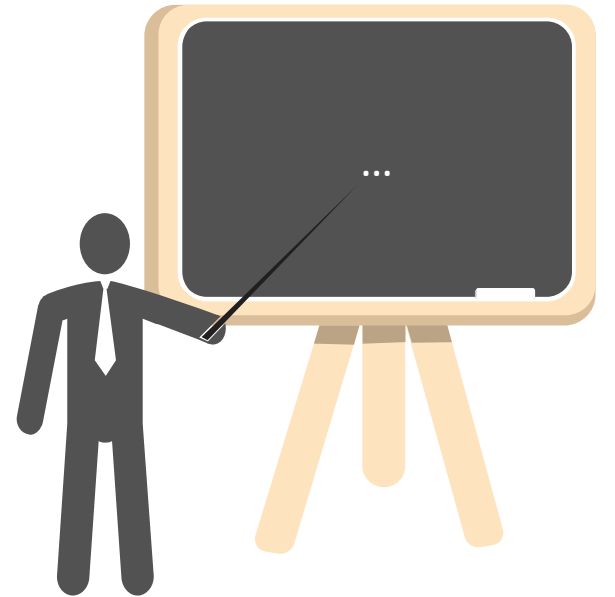


Traffic Surges During Double 11 or Other Big Promotions

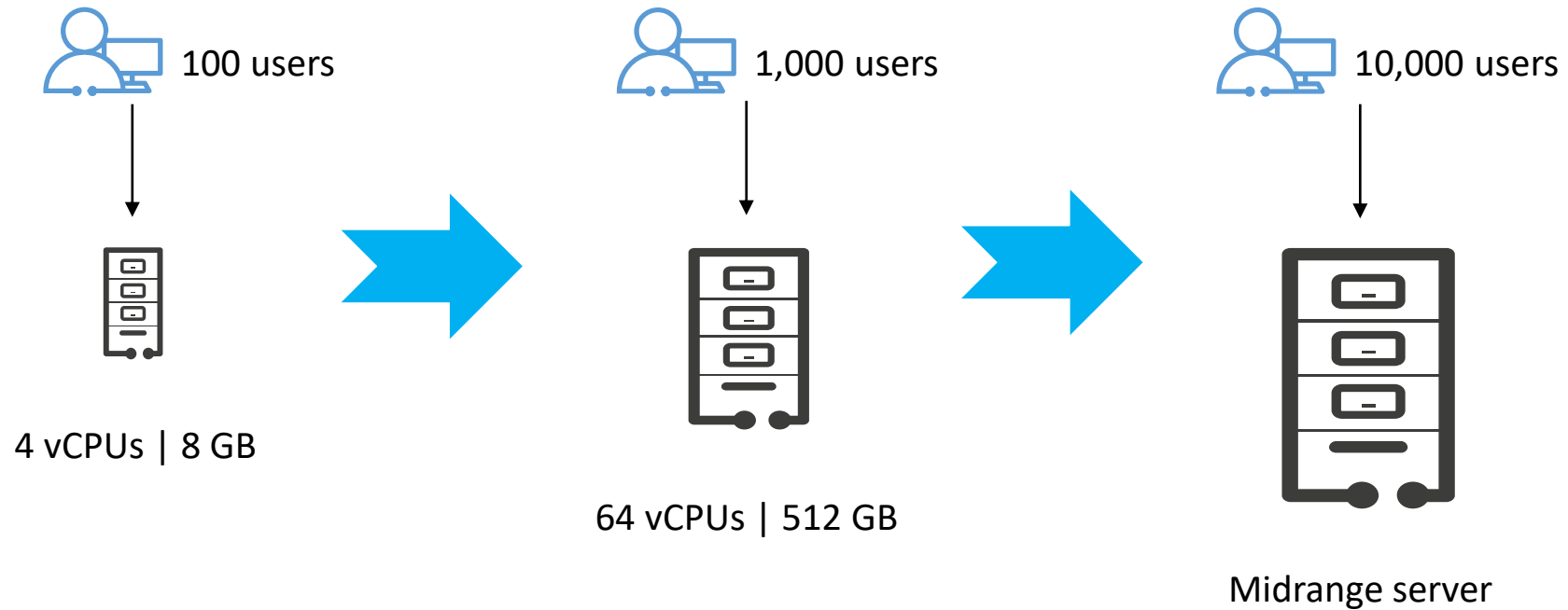


It is not just a waste of resources.

How Can We Manage Sudden Traffic Changes?



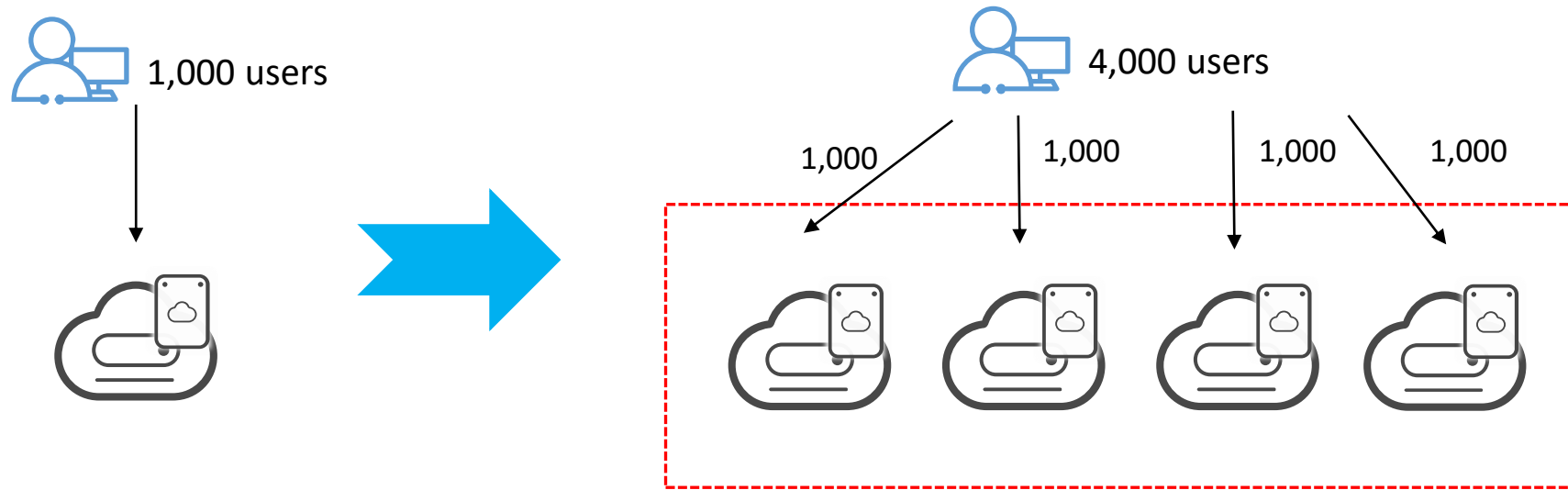
Method 1: Scaling Up



Advantages: simple and independent

Disadvantages: **more bottlenecks**

Method 2: Scaling Out



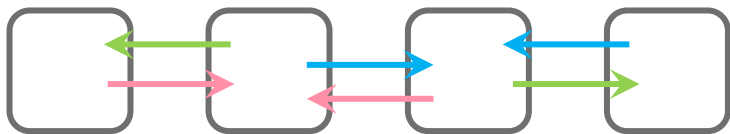
Advantages: You can use simple devices for unlimited scalability and higher availability.

Disadvantages: **Technology coordination is required, and this method only works for stateless applications**

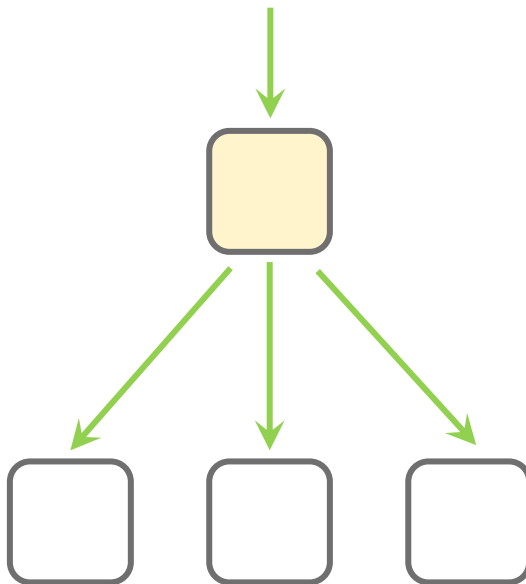


Distributed deployment seems better.

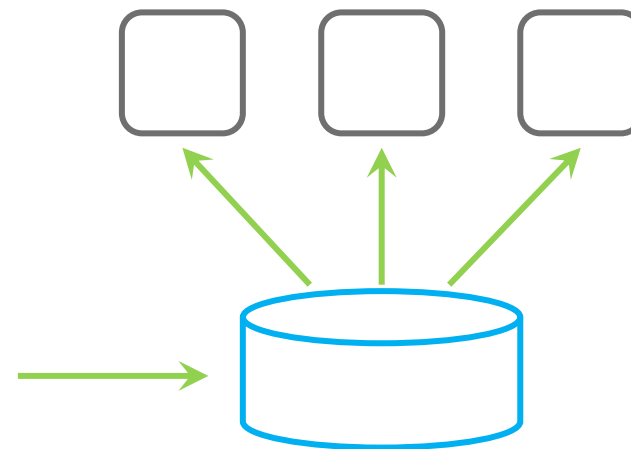
Typical Distributed Models



Synchronous communication and coordination

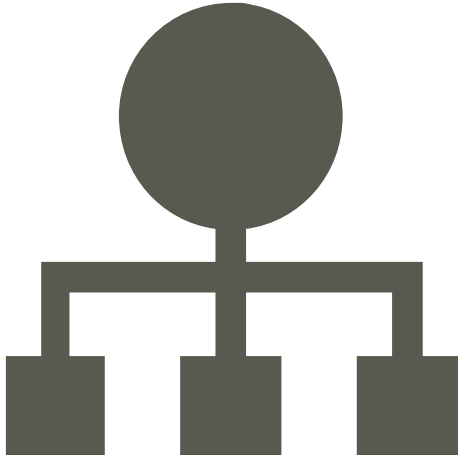


Request distribution

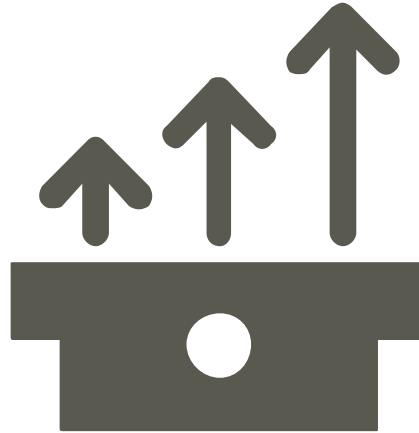


Task polling

Benefits of Distributed Systems



**Huge resource pool for
required performance**

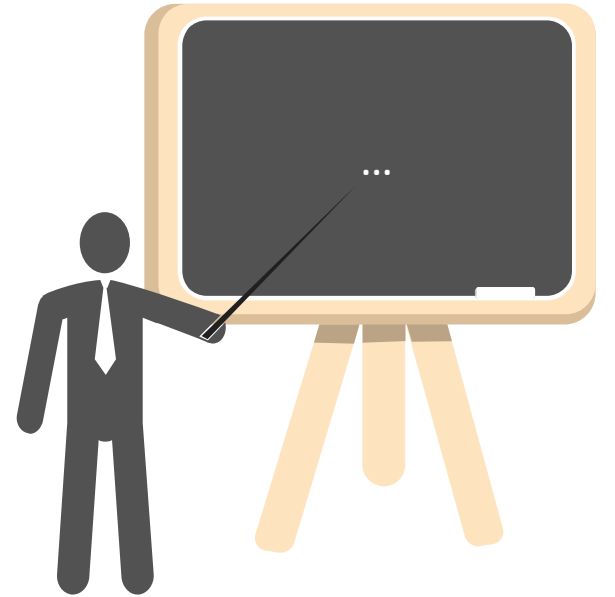


**Robust load balancing for
stable performance**

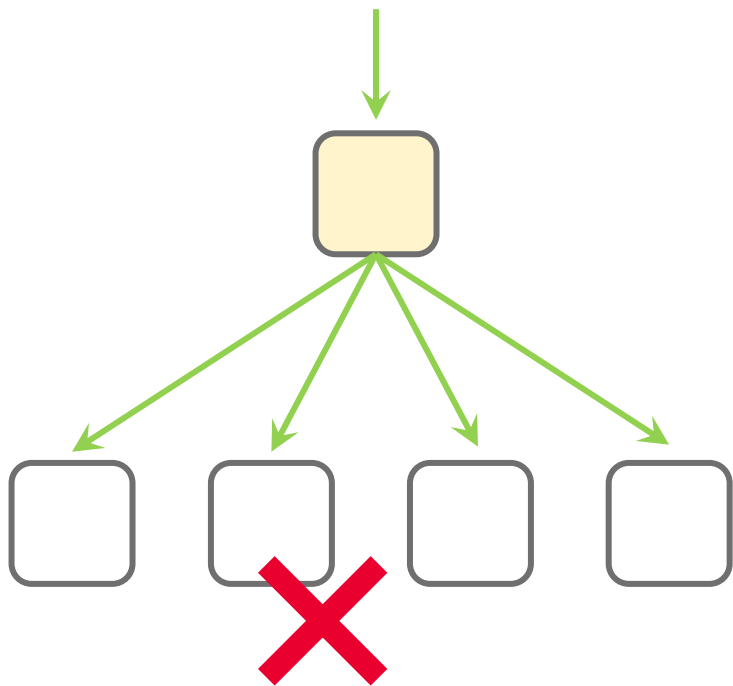


Better fault recovery

What Else Can We Get from Scalable Systems?

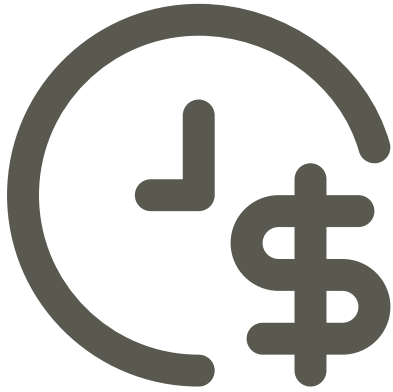


A Good Scalability Design Means Improved Reliability



- ❑ If a node is unhealthy, the overall processing capacity is reduced.
- ❑ If this happens, the system will automatically add a healthy backend server to ensure stable performance.

The Value of Scalability



Adjust resources as demand changes to reduce costs.



Replace faulty nodes with healthy ones for higher reliability.



Contents

1. Scalability Implementation
- 2. How to Build Scalable Applications**
3. Typical Scalable Website Architecture

What Are Needed to Enable Scalability?

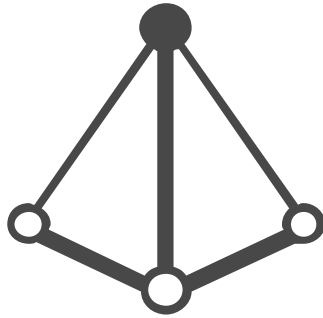
Step 1 Distribute traffic to multiple servers.

Step 2 Get to know when it is the time to scale resources.

Step 3 Scale resources immediately as requirements change.

Elastic Load Balance (ELB)

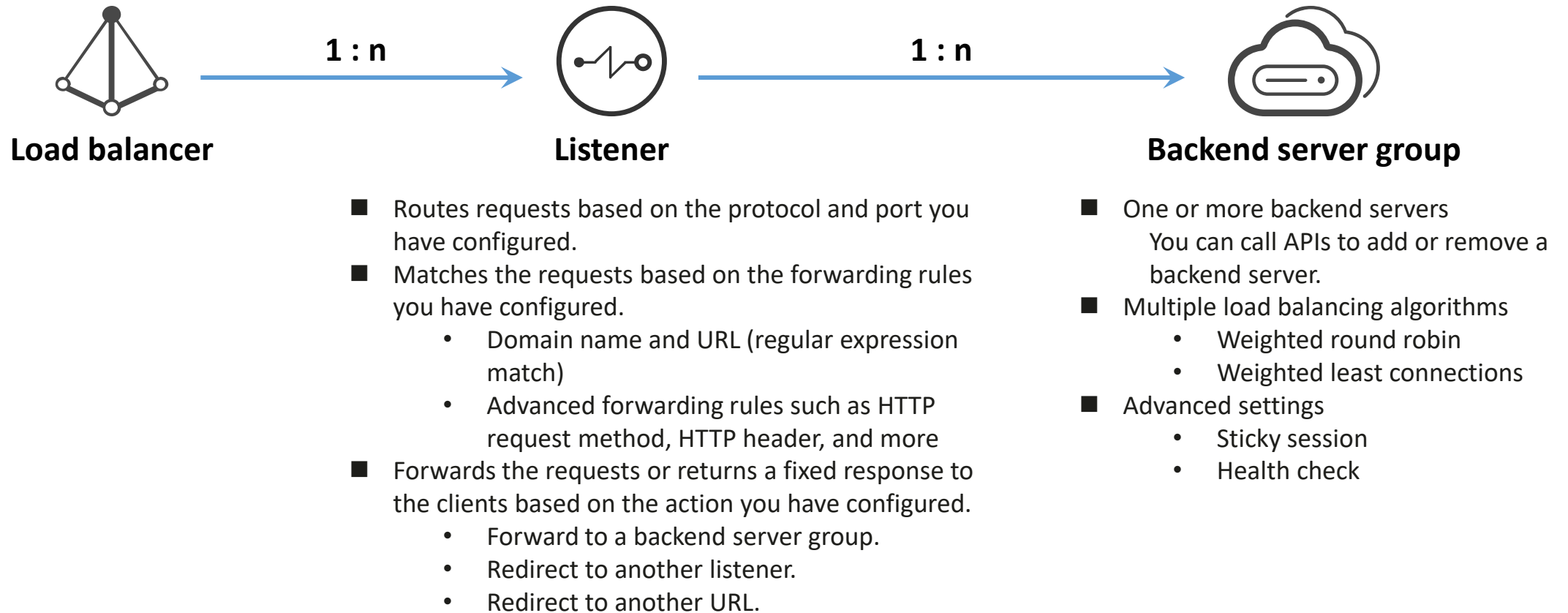
- ELB automatically distributes incoming traffic across multiple backend servers based on the listening rules you configure. It expands the service capabilities of your applications and improves their availability by eliminating single points of failure (SPOFs).



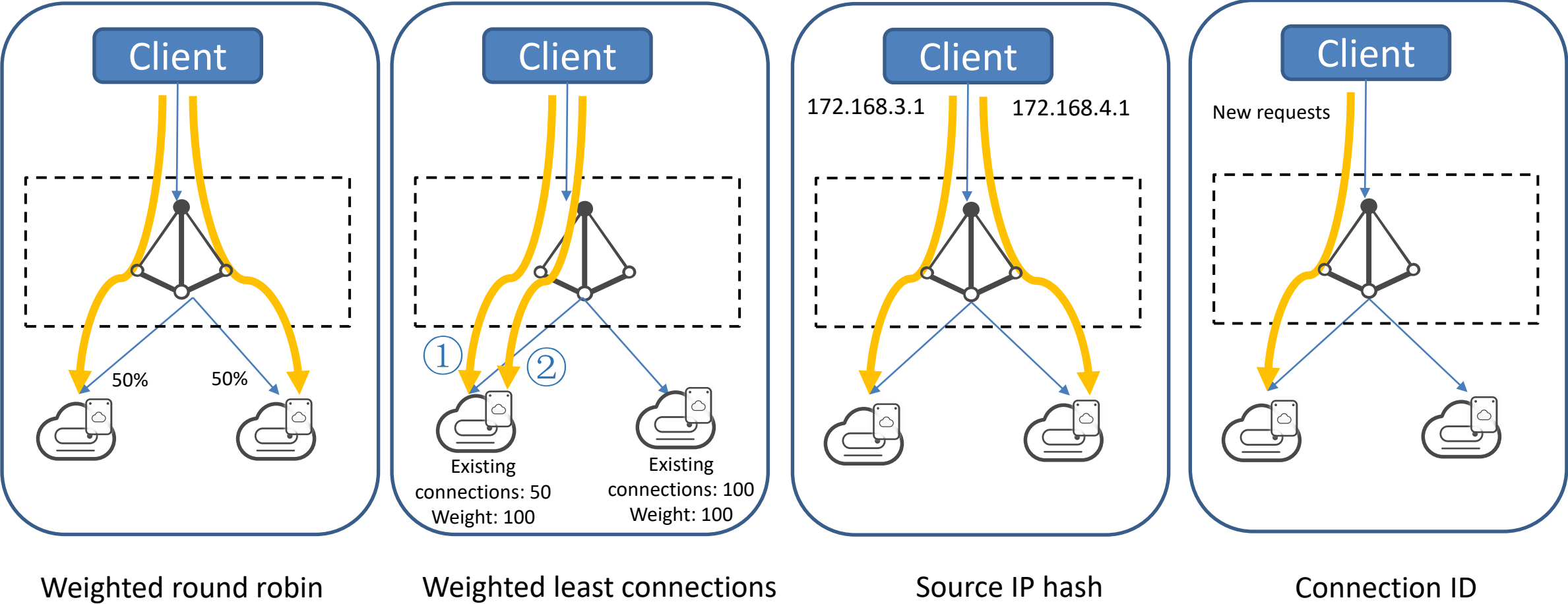
ELB

- ☐ Distributes requests across different backend servers.
- ☐ Supports built-in HA.
- ☐ Works with Auto Scaling to process a massive number of concurrent requests.
- ☐ Routes traffic across backend server groups based on the forwarding policies you have configured.
- ☐ Checks the health of backend servers to ensure that requests are routed to healthy servers.
- ☐ Supports load balancing at both Layer 4 and Layer 7.

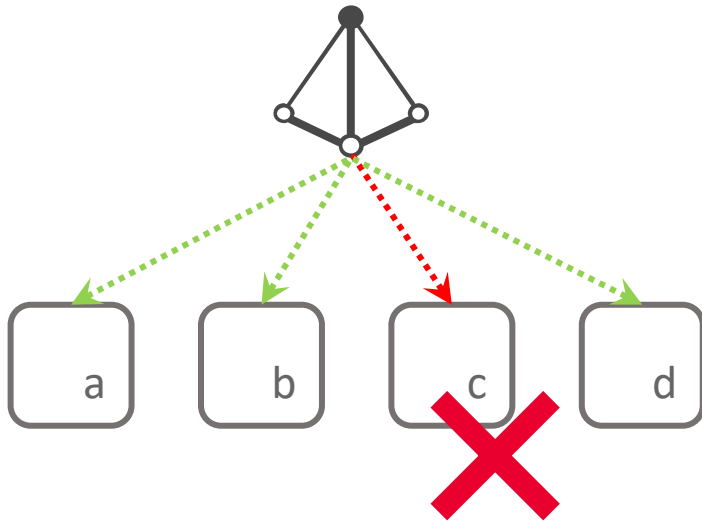
ELB Components



Load Balancing Algorithms



Health Check



- Configure a health check to avoid routing requests to unhealthy backend servers.
- How does a health check work?
 - ELB connects to a specific TCP port used by the backend server.
A connection is established with the port.
 - ELB accesses a specific HTTP page, generally the root directory.
A 2xx code is returned.
- You should select a dedicated page for health checks.

ELB Configuration Process

1. Creating a Load Balancer

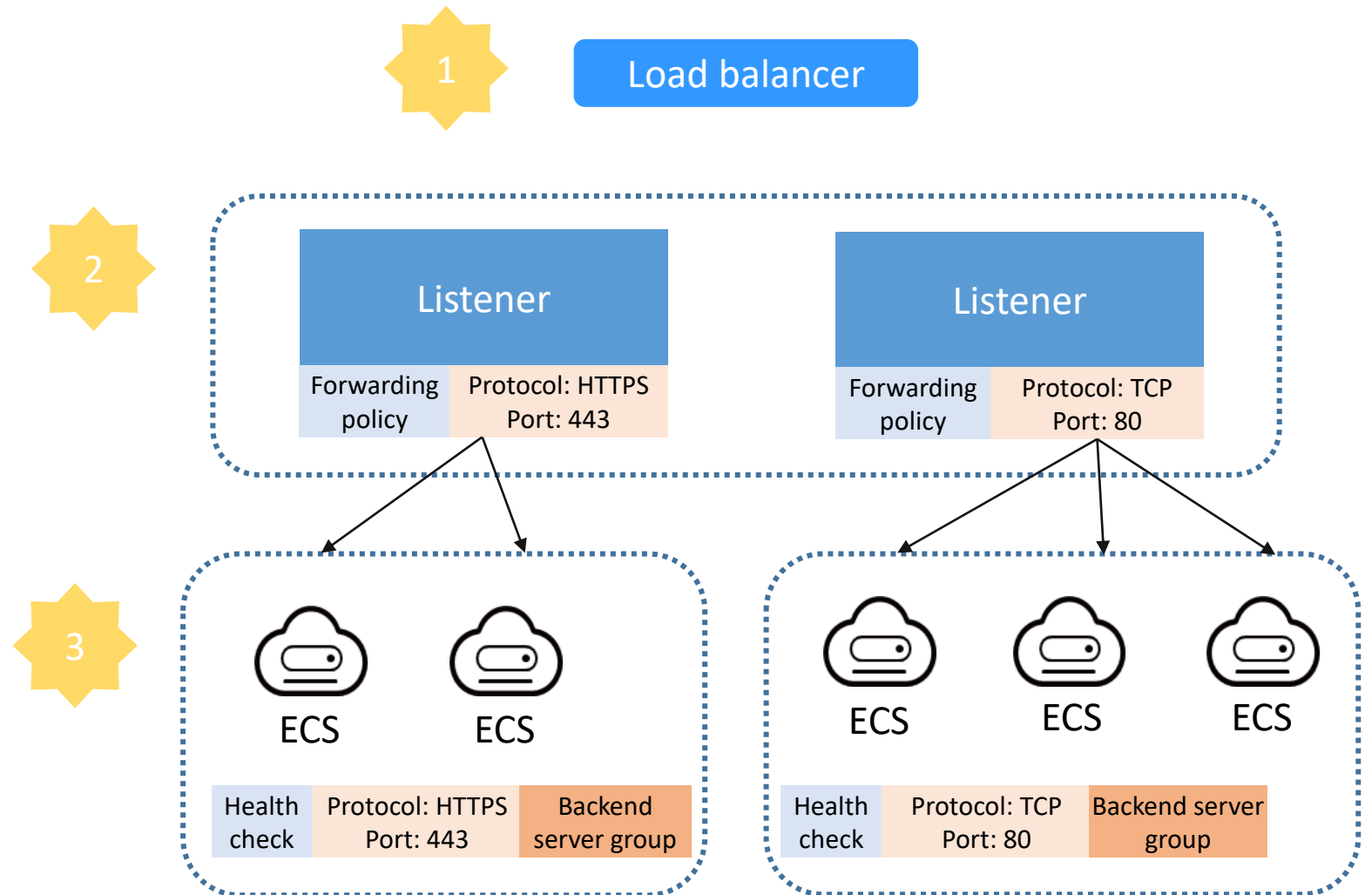
- ❑ Create a load balancer.
- ❑ Select the load balancer type.
- ❑ Configure the network parameters.

2. Adding a Listener

- ❑ Locate the created load balancer.
- ❑ Configure the protocol and port.

3. Adding a Backend Server Group

- ❑ Select a load balancing algorithm.
- ❑ Configure a health check.



Creating a Load Balancer

Basic Information

Type



Dedicated load balancer

They work well for heavy-traffic and highly concurrent services, such as large websites, cloud-native applications, IoT, and multi-AZ disaster recovery applications.



Shared load balancer

They are good for services with low traffic, such as small websites and common HA applications.

The load balancer type cannot be changed after it is selected. View [Differences Between Dedicated and Shared Load Balancers](#) before selecting a type.

Billing Mode

Yearly/Monthly **Pay-per-use**

Region

📍 CN East-Shanghai2

AZ

A... x A... x

You can choose to deploy the load balancer in multiple AZs for higher availability.

Name

elb-2d2a

Specification

Specifications

Fixed Elastic

For stable traffic

☒ Application load balancing(HTTP/HTTPS) ?

Small I | 2,000 new HTTP connections / 200 new HTTPS connections | 200,000 concurrent connections | 4,000 queries per ... [Compare specifications](#)

☒ Network load balancing(TCP/UDP) ?

Small I | 10,000 TCP / 10,000 UDP new connections | 500,000 TCP / 500,000 UDP concurrent connections | 50 Mbit/s of ba... [Compare specifications](#)

Network Configuration

Network Type

☒ Private IPv4 network

VPC

vpc-default [View VPCs](#) [Create VPC](#)

Once the load balancer is created, the VPC cannot be changed.

Frontend Subnet ?

--Select-- [View Subnet](#) [Create Subnet](#)

Backend Subnet ?

Subnet of the load balancer [View Subnet](#) [Create Subnet](#)

The load balancer requires a minimum of 12 IP addresses in the subnet.

Make sure that the security group and network ACL rules allow traffic from the backend subnet where the load balancer works to the backend servers.

[Learn how to configure a security group](#) [Configure Security Group Rule](#) [Configure Network ACL Rule](#)

Adding a Listener

1 Configure Listener

2 Configure Routing Policy

3 Add Backend Server

4 Confirm

★ Name

listener-b072

Frontend Protocol

The protocol used by the load balancer to receive requests from the clients. Select TCP, UDP for listeners at Layer 4, and select HTTP, HTTPS for listeners at Layer 7.

TCP

UDP

HTTP

HTTPS

TCP or UDP listeners do not support access logging.

★ Frontend Port

80

Value range: 1 to 65535

Access Control

All IP addresses

?

Transfer Client IP Address

☐

?

If you enable this option, servers cannot be backend servers and clients at the same time.

Advanced Settings 

Idle Timeout (s)

300

Description

--

Adding a Backend Server Group and Enabling Health Check

Configure Listener

Configure Routing Policy

3 Add Backend Server

4 Confirm

Backend Servers

Add Backend Server

Backend Servers	Private IP Address	Backend Port ?
No data available.		

Health Check

Health Check

Health check detects the running of backend servers and ensures that requests are routed only to healthy backend servers

Advanced Settings

Health Check Protocol	TCP	Health Check Port	Default backend server port
Interval (s)	5	Timeout (s)	3
Healthy Threshold	3	Unhealthy Threshold	3

Configure Health Check

Health Check

Health check detects the running of backend servers and ensures that requests are routed only to healthy backend servers.[Learn more](#)

The backend server group is associated with a dedicated load balancer. Security group rules need to be configured to allow traffic from the backend subnet where the load balancer is running to backend servers. If security group rules are not configured, backend servers cannot receive requests from the load balancer and will be identified as unhealthy.[Learn how to configure a security group.](#)

★ Health Check Protocol ?

HTTP

Domain Name

Private IP address of the backend server

☒ Specified domain name

www.example.com

Enter at least two character strings separated by periods (.). Use only letters, digits, and hyphens (-). Do not start or end strings with a hyphen. Max total: 100 characters. Max string: 63 characters.

Health Check Port

☒ Default backend server port

☐ Specified port

★ Path ?

/

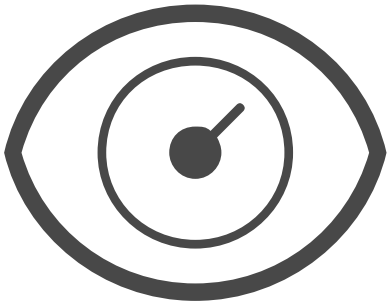
Start the path with a slash (/). The path can contain 1 to 80 characters, including letters, digits, and the following characters: / % ? &

OK

Cancel

Cloud Eye

- Cloud Eye is a multi-dimensional resource monitoring service. Users can use Cloud Eye to monitor resources, configure alarm rules, identify resource exceptions, and respond to resource changes.

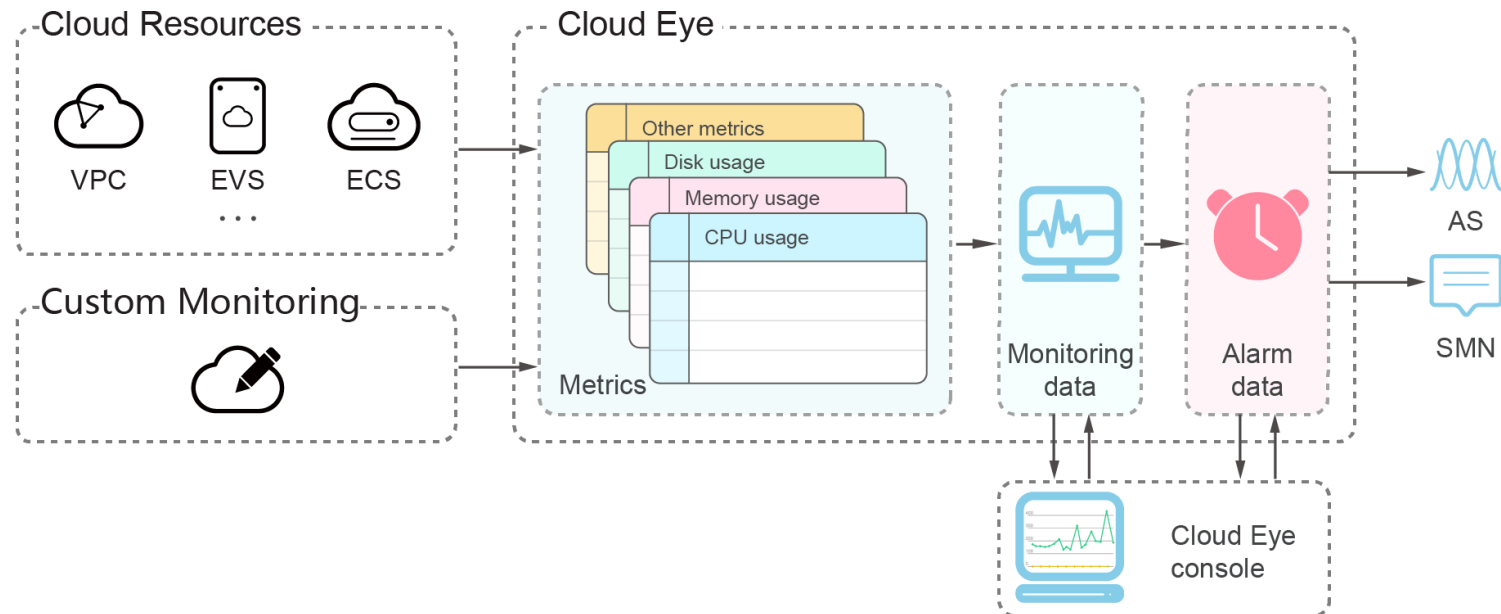


Cloud Eye

- ❑ O&M personnel cannot operate and maintain a system just based on their feelings.
- ❑ Cloud Eye is not a console where users can control different cloud services.
 - It is essentially a data collection system.
 - It is not tightly coupled with other services.
- ❑ Cloud Eye APIs can be integrated into third-party applications.
- ❑ Cloud Eye provides dashboards to visualize metrics and supports comprehensive alarm management.

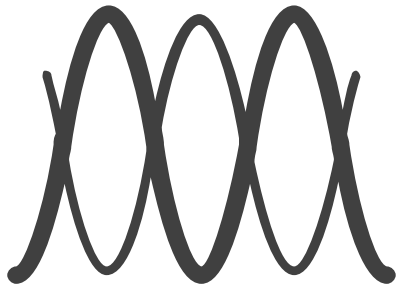
Cloud Eye Architecture

- Cloud Eye collects cloud resource or custom metrics in real time. It allows users to flexibly configure alarm rules based on the collected data. When alarms are triggered, Cloud Eye notifies users using methods such as email or SMS message, and some service systems can respond to alarms automatically to ensure that services run smoothly.



Auto Scaling

- Auto Scaling helps you automatically adjust Elastic Cloud Server (ECS) and bandwidth resources to keep up with changes in demand based on pre-configured scaling policies. It allows you to add ECS instances or bandwidth resources to handle increases in load and also save money by removing resources that are sitting idle.



Auto Scaling

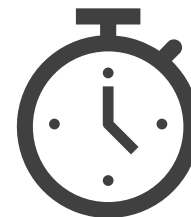
- ❑ Can create or delete ECS instances across AZs.
- ❑ Supports a broad range of conditions and policies to create or delete instances.
- ❑ Automatically identifies and replaces unhealthy instances.
- ❑ Works together with ELB to distribute incoming traffic across healthy instances.

Scaling Policies and Scenarios



-
- Real-time monitoring of performance
 - Automatic, dynamic scaling
 - Reactive scaling

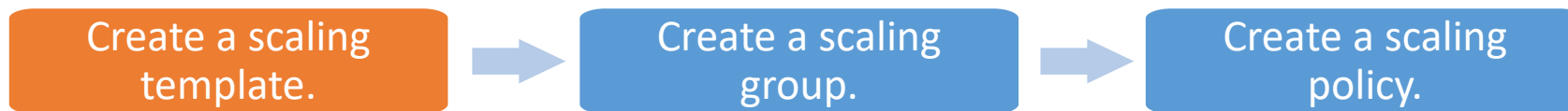
Dynamic scaling based on performance



-
- Regular workload changes
 - Scaling by schedule
 - Proactive scaling

**Scheduled scaling based on
predictable workload changes**

Using Auto Scaling - Creating a Scaling Template



The screenshot shows the 'Create a scaling template' form. It includes fields for Billing Mode (Pay-per-use), Region (CN South-Guangzhou), Name (as-config-2987), and Configuration Template (Create new template). Below these, there are tabs for CPU Architecture (x86, Kunpeng) and Specifications (General computing, General computing-plus, Memory-optimized, Large-memory, High-performance computing, Disk-intensive). A table lists ECS types and their specifications.

ECS Type	Flavor Name	vCPUs Memory ...	CPU	Assured / Maximum Bandwidth
☐ General computing s7	s7.small.1 (Sold out i...	1 vCPUs 1 GiB	Intel Ice Lake	0.1/0.8 Gbit/s
☐ General computing s7	s7.medium.2 (Sold ou...	1 vCPUs 2 GiB	Intel Ice Lake	0.1/0.8 Gbit/s

Creating a template from scratch

The screenshot shows the 'Create a scaling template' form with the 'Use existing ECS' option selected. It includes fields for Region (CN South-Guangzhou), Name (as-config-2987), Configuration Template (Use existing ECS), EIP (Do not use), Login Mode (Key pair), and Key Pair (Create Key Pair).

Creating a template from an existing ECS

Using Auto Scaling - Creating a Scaling Group

Create a scaling template.



Create a scaling group.



Create a scaling policy.

★ Region: CN South-Guangzhou

Regions are geographic areas isolated from each other. Resources are region-specific and cannot be used across region network latency and quick resource access, select the nearest region.

★ AZ: AZ7 × AZ6 × AZ5 × AZ3 × AZ2 × AZ1 ×

★ Multi-AZ Scaling Policy: ☒ Balanced ☐ Sequenced

★ Name: as-group-20d9

★ Max. Instances: 1

★ Expected Instances: 0

★ Min. Instances: 0

The selected AS configuration serves as a specifications template for the instances in your AS group. After a subnet is selected, an IP address will be automatically assigned.

★ AS Configuration: [Redacted] +

★ VPC: vpc-f298 (192.168.0.0/16) Create VPC

★ Subnet: subnet-f328 (192.168.0.0/24) This subnet is used by the primary NIC.

☒ Source/Destination Check

+ Add Subnet You can add 4 more subnets. Create Subnet

Load Balancing: Do not use Elastic load balancer

Load Balancing: Do not use Elastic load balancer

★ Instance Removal Policy: Oldest instance created from oldest AS conf...

EIP: Release Do not release

Data Disk: Delete Do not delete

★ Health Check Method: ECS health check

If a protected instance is identified as unhealthy in a health check, AS replaces the instance.

★ Health Check Interval: 5 minutes

★ Health Check Grace Period (s): 600

Tag: Tag key Tag value

You can add 10 more tags.

Agency: --Select-- Create Agency

★ Agreement: ☐ I have read and agree to the Auto Scaling Disclaimer.

Using Auto Scaling - Creating a Scaling Policy

Create a scaling
template.

Create a scaling
group.

Create a scaling
policy.

Add AS Policy

Policy Name:

Policy Type: **Alarm** | Scheduled | Periodic
Policies of this type are applied only when their associated alarm rules are enabled. [View alarm rules](#)

Alarm Rule: **Create** | Use existing

Rule Name:

Monitoring Type: **System monitoring** | Custom monitoring

Trigger Condition: CPU Usage | Max. | > | %
If you select a metric starting with (Agent), the Agent must be installed on all instances in the AS group. [Learn more](#)
For more information about AS monitoring metrics, see [Monitoring Metrics](#).
The metrics that can be monitored vary somewhat by OS. [Learn more](#)

Monitoring Interval:

Consecutive Occurrences: ?

Alarm Policy Type: **Simplified scaling** | Refined scaling

Scaling Action: Add | 1 | instances

Cooldown Period (s): ?

Alarm-based scaling policy

Add AS Policy

Policy Name:

Policy Type: Alarm | **Scheduled** | Periodic

Time Zone: GMT+08:00

Triggered On: ?
The specified time must be later than the default time and the current system time.

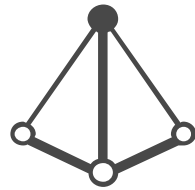
Scaling Action: Add | 1 | instances

Cooldown Period (s): ?

Scheduled scaling policy

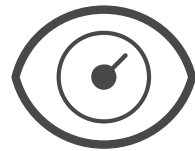
Cloud Services for Creating Scalable Systems

Step 1 Distribute traffic to multiple servers.



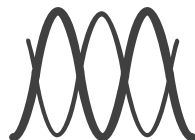
ELB

Step 2 Get to know when it is the time to scale resources.



Cloud Eye

Step 3 Quickly scale in or out resources.



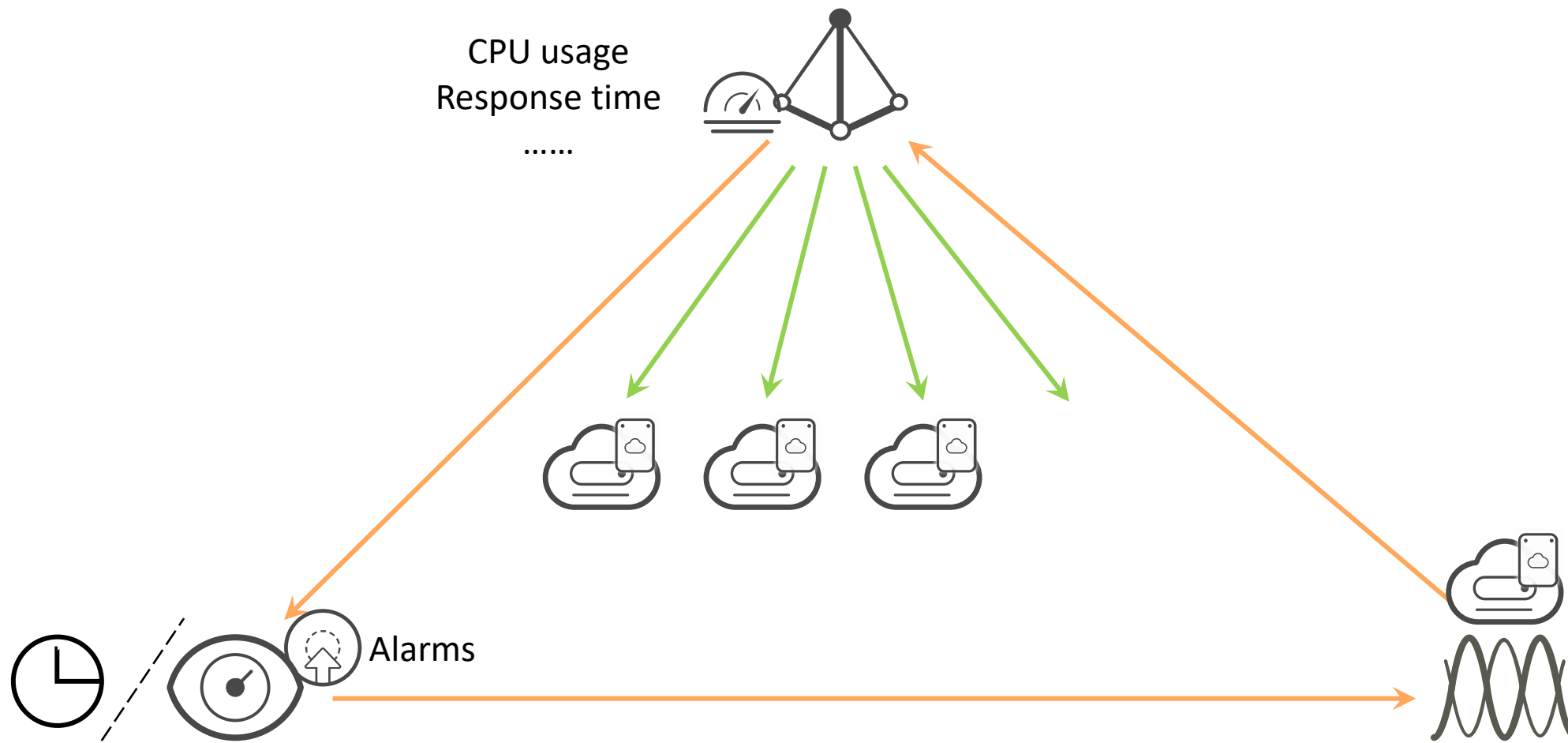
Auto Scaling



Contents

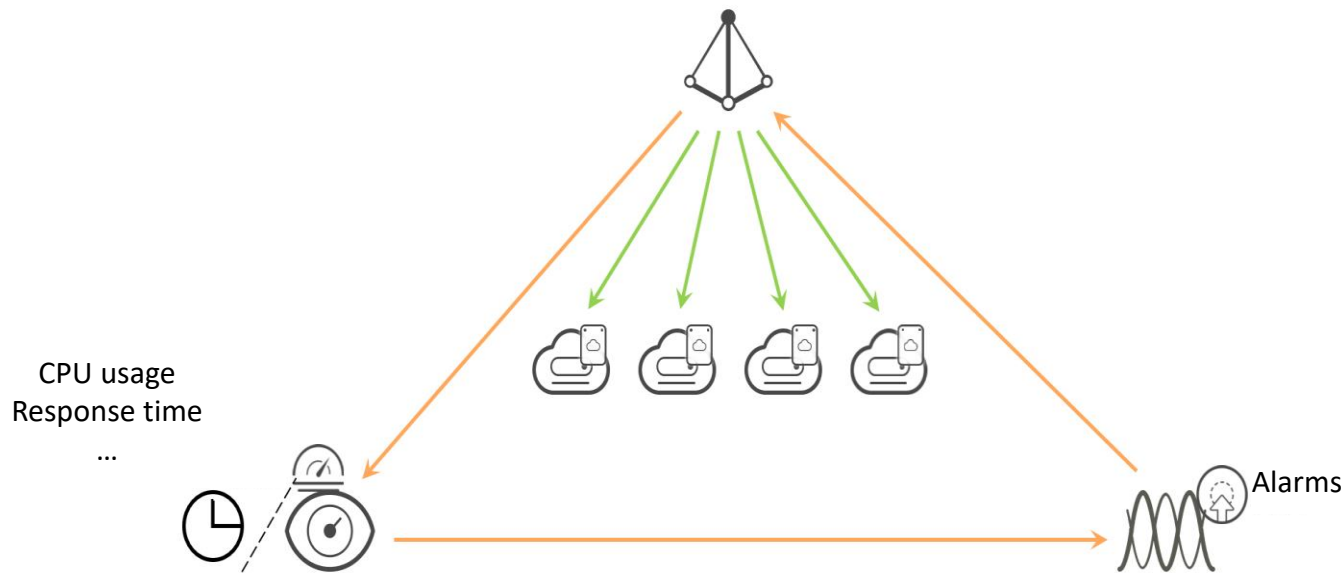
1. Scalability Implementation
2. How to Build Scalable Applications
- 3. Typical Scalable Website Architecture**

Typical Scalable Website Architecture



Application Scenarios

- **Heavy-traffic forums:** Service load changes are difficult to predict for heavy-traffic forum websites. AS dynamically adjusts the number of ECSs based on monitored ECS metrics, such as CPU usage and memory usage.
- **Livestreaming:** A livestreaming website may broadcast popular programs from 14:00 to 16:00 every day. Auto Scaling automatically scales out ECS and bandwidth resources during this period to ensure a smooth viewer experience.



- **E-commerce:** During big promotions, E-commerce websites need more resources. Auto Scaling automatically add resources within minutes to ensure that promotions go smoothly.



Summary

- Learned the advantages of load balancing on the cloud.
- Learned how to schedule distributed applications.
- Learned how to adjust resources for distributed applications.

Quiz

For a scalable system on Huawei Cloud, which of the following can be used to create or delete servers? ()

- A. Elastic Load Balance (ELB)
- B. Cloud Eye
- C. Auto Scaling
- D. Elastic Cloud Server (ECS)



Acronyms and Abbreviations

- ECS: Elastic Cloud Server
- EVS: Elastic Volume Service



Recommendations

- Huawei Cloud websites
 - Huawei Cloud: <https://www.huaweicloud.com/intl/en-us/>
 - Huawei Cloud Developer Institute: <https://edu.huaweicloud.com/intl/en-us/>



Huawei Cloud
Developer Institute

Thank You.

Copyright © 2024 Huawei Technologies Co., Ltd. All Rights Reserved.

The information in this document may contain predictive statements including, without limitation, statements regarding the future financial and operating results, future product portfolio, new technology, etc. There are a number of factors that could cause actual results and developments to differ materially from those expressed or implied in the predictive statements. Therefore, such information is provided for reference purpose only and constitutes neither an offer nor an acceptance. Huawei may change the information at any time without notice.



www.huaweicloud.com